

SPICES

THEIR HISTORIES

Valuable Information For Grocers



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Title: Spices, Their Histories: Valuable Information for Grocers

Author: Robert O. Fielding

Release date: August 29, 2019 [eBook #60192]

Most recently updated: October 17, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/60192

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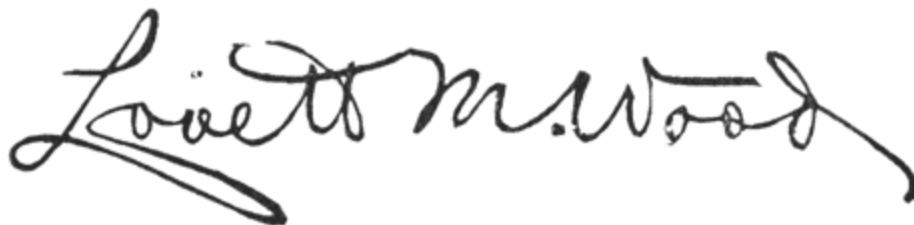
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SPICES

INTRODUCTION.

The history of spices, with other valuable information to all branches of the grocery trade, was originally written by Robert O. Fielding, of the staff of the Trade Register, in which the several articles appeared in various issues of that journal, duly protected by copyright, with the accompanying illustrations.

Retail grocers everywhere will find this little book of especial value for study and reference. It is all meat for the salesman who realizes that success in trade these days depends upon knowing where the goods he handles were produced, how to judge their qualities, how they are prepared for market, and what are their uses. How to sell, the market conditions, etc., are continuously set forth in the weekly issues of the Trade Register, \$2 a year, by men who have had practical experience behind the counter.

A handwritten signature in cursive script, reading "Lovett M. Wood". The signature is fluid and elegant, with a long, sweeping underline that extends to the right.

Editor.

ALLSPICE OR PIMENTO

A Valuable Product From Jamaica Which Combines the Flavor of Cloves, Cinnamon and Nutmeg

Allspice is the dried unripe berries of a tree of the myrtle family, the pimento, known botanically as *Eugenia pimenta*, or *Pimenta officinalis*. It's an evergreen tree some 20 to 30 ft. high, with a slender, straight, upright trunk, much branched at the top; the bark is smooth, gray and aromatic; the leaves—which when fresh abound in essential oil—are 5 in. long, of an oblong shape and deep shiny green color; the blossoms—which appear in July and August—are white and fragrant; the berries (sometimes called corns), which form on the disappearance of the flower, are picked unripe, altho fully grown, they are of a greenish-purple color. After picking, the berries are dried in the sun or in kilns until dark brown and then separated from the stalk. The dried berries are light, brittle, of roundish form and crowned with the remains of the flower calyx in the shape of a raised, seared-like ring; each berry contains two dark-brown flattish, kidney-shaped seeds. If allowed to ripen, the berries lose their aromatic flavor and become merely sweet and pulpy. Only in Jamaica—where it is cultivated in plantations called Pimento walks—does the pimento tree grow to perfection, altho attempts are made to cultivate it in other West India islands and South America. It is thought to combine the flavor of cloves, cinnamon and nutmeg, hence it is called allspice.



Uses—Its chief use is for culinary purposes. It is a powerful irritant, good for dyspepsia, flatulency, gout, hysteria and toothache. It is often employed to disguise the nasty taste of medicine. Allspice yields volatile oil by distillation, which is used as a flavoring in alcoholic solution, is of a brownish-red, clear appearance, and has the odor and taste of pimento, but is warm and more pungent. A green fixed oil has the burning aromatic taste of pimento and is supposed to be the acrid principle. A tincture from allspice has been praised as an application in chilblains.

Substitutes.—The Mexican spice, called Pimento de Tabasco is somewhat larger and less aromatic than Jamaica pimento. The berries of

Pimento acris, (bayberry) whose leaves are used in the manufacture of bay-rum. The Carolina allspice—*calycanthus florides*, a shrub 6 or 8 ft. high, with an odor somewhat like strawberries. Japan allspice—*chimonanthus fragrans*—which grows in Japan, and wild allspice—*lindera benzoin*—known also as spice-wood, fever-wood, benjamin-bush—a member of the laurel family growing in the United States. To secure uniformity of color these inferior kinds are often colored with Armenian bole, a kind of red clay from Armenia, and they are also often mixed in ground allspice, in addition to the stalks of the pimento. A kind of red pimento from Salonica is also used as an adulterant. During the civil war, when pimento was high in price, a substitute was made up of clove-stems, wasted rye, a little cayenne pepper, and some cassia; this was very acceptable, altho there was not an ounce of pimento in it.

CAPSICUM

Cayenne Pepper Is Made from This Branch of the Nightshade Family— Descriptions of the Various Varieties of Capsicum—Tabasco Pepper Sauce

The capsicum is a genus of plants of the nightshade family (Solanaceae) that grows luxuriantly in all tropical countries and many species of which are cultivated in the temperate zone. Capsicum or Red Pepper is of American origin for these reasons: Fruits so conspicuous, so easily grown in gardens and so agreeable to the palates of the inhabitants of hot countries would have very quickly diffused throughout the old world, if they had existed in the South of India, as it has sometimes been supposed. They would have had names in several ancient languages, yet neither the Romans, Greeks nor the Hebrews were acquainted with them. They are not mentioned in ancient classic books. The islands of the Pacific did not cultivate them at the time of Cook's voyage in spite of the proximity of the Sunda Isle where Rumphius mentions their very general use. The Arabian physician, Ebn Baithar, who collected in the 13th century all that eastern nations knew about medicinal plants, says nothing about them. Probably the first known history of cayenne pepper in Europe is that given by Martyr, who writes of Columbus bringing it home in 1493 and speaks of it being more pungent than that from Caucasus, probably referring to the Oriental black pepper. About a century later, Gerard writes of its being brought into Europe from Africa and Southern Asia and being grown in European gardens. Probably the first record of its use is that by Dr. Chanca, who was physician with Columbus' fleet in 1494, and who alludes to it as a condiment used in dressing meats,

dying and other purposes, as well as a medicine. From the ground dried seeds and pericarp of certain varieties of capsicum we get cayenne pepper, so-called from Cayenne, in French Guiana, S. A., whence it was imported by the French. Cayenne pepper is also called Calicut and Napaul, the names of places of export, and it was known as Guiana pepper over 300 years ago. The derivation of the word “Capsicum” is uncertain; it may be from Kapto, hot, on account of its pungent taste, or from capsa, a box, or chest, referring to the form of its fruit. The plant grows from 1 ft. to 6 ft. high and is fairly well branched; the flowers are white or greenish-white; the fruits of the several species are of various forms, round, oblong, cordate or horned, and contain a number of flattish seeds. The seeds after the removal of the pericarp, and then thoroly washed and dried, are entirely devoid of acidity and pungency. The hotter and drier the soil, the more acrid and pungent the fruit.



Used in moderate quantities, capsicum or cayenne pepper, promotes digestion and so prevents flatulence. The natives of Brazil boil the capsicums and dip their manioc bread in it, making a kind of fiery soup. They are extensively used in India in compounding curries and chutneys. In Bengal the natives make an extract from the small capsicum chilies of about the consistency of molasses. The bell peppers are pleasant stuffed with meats, fish, other vegetables, etc. The sweet and mild kinds fed to birds are said to improve their plumage.

C. Annum is the most common species and contains a great many varieties, among them the Pimiento (not Pimento or allspice) commonly known as Spanish red peppers or morrons, also Paprika, or Hungarian sweet pepper. This species is never found growing wild.

C. frutescens is sometimes called goat pepper and is generally described as the true cayenne. Its leaves are from 3 in. to 6 in. long by 2 in. to 3½ in. wide, the fruit is red, obtuse or oblong acuminate, ¾ to 1½ in. long and ¼ to ¾ in. in diameter. It is very acrid and pungent. It is only cultivated in the tropical regions, as the seasons in the temperate climate are not long enough to mature the fruit.

C. baccatum is ovate or sub-round and about ¼ in. in diameter. *C. baccatum* have been known in the English gardens since 1731.

C. fasciculatum, also known as Mexican chilies, is a shrubby plant of Sierra Leone, and grows in Zanzibar; also known as small chilies, or red cluster peppers. The fruit, which grows erect, is oblong linear, not quite an inch in length and of a deep red orange color. Another variety, which are mostly consumed locally, have larger red and yellow fruit. Zanzibar capsicums or chilies, are dirty looking, of a brownish-red color and very hot. A variety from Japan are bright red, not so pungent as the other growths, but of finer aroma.

C. cerasiforme, the fruit is spherical, sub-cordate, oblate or occasionally pointed. The flesh is firm, from 1-12 to ½ in. thick, and very pungent; from the shape of its fruit it is called the cherry capsicum, or pepper.

C. grossum, originally from India, grows 2 ft. high, with a few branches and large leaves 3 to 5 in. long, the fruit is large, oblong or ovate, and is known as bell pepper; it is mostly used for stuffing and pickling; the skin being thick, soft and tender and of a mild flavor.

C. abbotii, with ovate fruit about 2 in. long. While this variety is used to some extent for pickling, it is cultivated more as an ornamental plant.

C. longum grows to about 3 ft. high with comparatively few branches, the fruit is often a foot long and 2 in. in diameter. The flesh is thick and flavor mild.

C. Acumination is about 2½ ft. high. The fruit, which is small, grows both erect and pendent.

C. Conordes, with oblong linear fruit, which grows erect, is very acrid and pungent. It is known as tabasco capsicum or pepper. Bird pepper, bird's-eye chilies, red-bird pepper, etc., are commercial names given to the mild, sweet varieties of capsicum on account of their being fed to birds. Nepaul pepper, commercial name for capsicum imported from that place in India. Nepaul pepper has an odor and flavor resembling orris root and a pod the color of amber when dried. It is most esteemed as a condiment, being aromatic and appetizing, and not so acrid or biting as is most cayenne. Paprika, commercial name for the mild, sweet varieties of capsicum, chiefly grown in Hungary, Spain, Portugal, Jamaica, Japan and Zanzibar.

Japanese pepper is the fruit of *Xanthoxylum*, an entirely different genus of plants to the capsicum family. The fruit capsules when bruised are agreeably pungent and aromatic. It is much esteemed as a condiment in China and Japan.

Tabasco pepper sauce originated with Mr. E. McIlhenny, of New Iberia, La., in 1868, from a variety of capsicum in which the fruit grows erect, and was brot by a soldier friend of Mr. McIlhenny from Tabasco in Mexico after the close of the Mexican war.

Tabasco catsup originated with Mr. George Bayle of St. Louis, Mo. The base of it is said to be equal proportions of powdered capsicum and essence of tomatoes.

Ground cayenne pepper soon loses its bright color when kept too long or exposed to the light, and becomes dingy in appearance, so it is not always wise to judge by looks alone, as red ocher, turmeric, mustard, rice, sawdust, salt, brick dust, etc., have been found in cayenne pepper.

The large fruits or pods are commercially known as capsicums, and the smaller ones as chilies. The term pepper is a misnomer as applied to this spice.

CINNAMON AND CASSIA.

The Sweet Wood of Ceylon and the Aromatic Bark of the Present Day Often Confused With Cassia—Valuable Trade History.

Cinnamon

As in the case of sago and tapioca, a good deal of misconception prevails in regard to cinnamon and cassia, and as with sago and tapioca, one is often sold for the other by the uninformed. The word “cassia,” botanically speaking, has nothing whatever to do with the aromatic bark which we call by that name, but refers to a genus of plants of the bean family, from which are derived the dried senna leaves, an infusion of which our mothers induced us to take by the bribe of a piece of candy, altho we had “tummy ache” for a brief space afterwards. The word “cinnamon” is derived from two Malayan words “cassia” from the Greek word “kasian,” which occurs in Psalms XLV-8, and elsewhere in the Bible, where it is supposed to refer to the aromatic bark of the present day, was afterwards tacked on. That cassia (the bark) was known in biblical times is well authenticated. It is mentioned in a Chinese herbal published in 1700 B. C. under the name kwei.

The earliest mention of cinnamon is in a list of offerings by Seleneneus Callinieus, king of Syria, and his brother, Antiochus Hierax, to the temple of Apollo at Miletus, 243 B. C. Among the gifts mentioned are: “2 lbs. of

cassia and a like quantity of cinnamon.” From this it appears there was then a recognized distinction between the two barks. We do know that the cassia was obtained from China, but the source of the cinnamon is unknown, unless it was obtained thru the Chinese from Ceylon, the inhabitants of those countries being in frequent intercourse in ancient times, for the earliest mention we have of cinnamon as a production of Ceylon is by Kazwini, an Arab writer of about 1275 A. D.



That cinnamon and cassia were extremely analogous is proved by the remark of the Greek physician Galen (130–200 A. D.): “The finest cassia

differs so little from the lowest quality of cinnamon, that the first may be substituted for the second, provided a double quantity of it were used.” With this brief historical sketch we will now endeavor to point out the differences between the two barks.

In the first place the word “cinnamon” refers solely to the cinnamon zeylanicum plant of Ceylon, where it is found growing wild, and was first brot under cultivation by De Koke in 1770. Here again, as with cloves, mace, etc., the Dutch tried to monopolize the trade. The giving away of a plant was punishable by flogging and the destruction of a plant involved the penalty of death. The tree grows to the height of 20 or 30 ft., having a trunk 12 to 18 inches in diameter; the leaves are of a thick leathery texture, 4 to 6 inches long, very smooth and shining on the upper surface, glaucous with prominent netted veins on the under side, and are traversed by 3 or 5 ribs. The flowers are greenish-white and appear in clusters of threes. The fruit is an oval berry, not unlike an acorn in shape and color. The tree flowers in January and the fruit ripens in August. When the branches are peeled the finest sticks are said to be derived from the liber of the middle-sized branches, an inferior sort from the younger shoots, and that which is procured from the thickest branches is considered of little worth. The peeling commences in May and lasts until November. The shoots or branches, usually about $\frac{1}{2}$ -inch to $\frac{3}{4}$ -inch in diameter and from 3 to 5 ft. long, are cut off with a curved pruning knife, tied up in bundles and carried to the peeling sheds. The bark is removed with a small, round-pointed knife, with a small projecting rib or cutter placed at right angles to the edge of the knife. With this knife the bark is split lengthwise of the stock. It is then carefully loosened from the wood for a short distance on either side of the slit. A similar incision is made on the opposite side and the bark is finally removed. The bark is then put in piles, covered with scrapings and matting and left for about two days, during which time a sort of fermentation takes place, which greatly facilitates the separation of the outer part of the bark from the cuticle and epidermis, which is carefully done by scraping with a small, curved knife, having a slightly serrated edge. This process is called piping. The piper sorts the bark as he scrapes it. He selects a slip suitable for the outer layer, about 3 ft. long, and packs within it 6 or 8 other pieces, all about the thickness of vellum paper—a mark which always distinguishes Ceylon cinnamon from cassia. They are then rolled up together and exposed to the sun to dry. It now resembles a tight roll of paper, the best quality

being firm and compact, of a golden yellow color, smooth on both outer and inner surfaces. The cheaper grades are not so carefully made, having many short pieces in the pipes or quills and not so much attention is paid to obtain uniform size and color. At Colombo it is sorted into three kinds by government inspectors. The two finest kinds are exported, the third with the broken pieces being reserved for obtaining oil of cinnamon. It is formed in bales about 90 lbs. each and wrapped in double cloths made of hemp, and not, as stated by some, of the cocoa tree.

Guava bark, soaked in the water left after the distillation of cinnamon oil and rubbed over with cinnamon oil, is sometimes placed inside good cinnamon quills and then it takes a man of Solomon's wisdom to detect the fraud.

Cassia

Cassia, under the name of Kwei, is mentioned in the earliest Chinese herbal—that of the Emperor Shena-ming, who reigned about 2700 B. C.; in the ancient Chinese classics, and in Rh-ya an herbal dating from 1200 B. C. In the Hai-yao-pen-ts'ao, written in the eighth century, mention is made of Tien-chu Kwei. Tien-chu is the ancient name for India, perhaps the allusion may be to the cassia bark of Malabar. In connection with these extremely early references to the spice, it may be stated that a bark supposed to be cassia is mentioned as imported into Egypt together with gold, ivory, frankincense, precious woods and apes, in the 17th century B. C. The accounts given by Dioscondes, Ptolemy and the author of the *Periphes of the Erythrean Sea*, that cinnamon and cassia were obtained from Arabia and eastern Africa; and we further know that the importers were Phoenicians who traded by Egypt and the Red Sea with Arabia, and it was imported hither from southern China.

Cassia, according to Marshall and others, is the bark of the old branches and trunks of the cinnamon *zeylanicum*, while others assert that it is the bark of an entirely different species, namely, cinnamon cassia, a native of China, but also grown in Java. This view is the more probable, as no cassia is exported from Ceylon, it almost all coming from Canton.

Regents have also very different effects on the infusion and oil of these two barks, which conclusively shows that they are obtained from different species. Cassia comes in bales, 2 to 4 lbs., bound by strips from the bark of some other tree. The pipes or quills are thicker and rolled once or twice, and never contain thinner pieces within; the diameter of the bark is much thicker, harder, and not as carefully scraped. The color is a deeper brownish-fawn color. The taste is more acridly aromatic, pungent sweet, at the same time more powerfully astringent yet muciliginous. Cassia is often substituted for cinnamon. It is adulterated with cassia lignea, the bark of a degenerate variety of cinnamon zeylanicum growing in Malabar, Penang and Silhet.

Other varieties of cassia are: Saigon cassia, the bark of an unknown species which appeared in commerce about 1875. The outer bark is not removed, has a gray or grayish-brown color, is covered externally with whitish blotches, warts or wrinkles.

C. Aromaticum is believed to be the cinnamon of China and Cochin China, growing in the provinces of Kwantung and Kwangsi. The leaves are very much larger than the Ceylon tree, hang down from the stalks and have never more than three ribs. This is the species that yields the cassia buds.

C. Tamala is a native of India, wild in Derwanee and Gongachora. It is cultivated in the gardens of Rungpoor. The dried leaves have an aromatic taste.

C. Loureirii grows in the lofty mountains of Cochin China, to the west towards Laos, Japan. The flowers of cassia are produced by this species. The old and young branches are worthless, but the middle-sized shoots produce a bark that is superior to that of Ceylon. None of it is exported.

C. Culilawan is a native of Amboyna. The bark when dry is aromatic like cloves, but less pungent and sweeter. It is used by the natives of Amboyna as an internal medicine and as a stimulating linament.

C. Rubrium grows in Cochin China, and contains an essential oil, smelling of cloves, but not so agreeable.

C. Sintoc is a tree about 80 ft. high, growing in the Neilgherry mountains, India, and the higher mountains of Java. The bark is of the same

quality as the Amboyna cassia, but not so agreeable. It is more bitter and powdery when chewed.

C. Xanthaneuron is a native of the Papuan islands and the Moluccas. The bark when fresh is very fragrant, but it soon loses its quality.

C. Nitidum is a native of India. It is a shrub or small tree.

C. Javanicum grows in Java and Borneo. It is a tree of about 20 to 30 ft. high. The dried bark is of a deep cinnamon brown color; more bitter than the Ceylon cinnamon, and the leaves when rubbed have a sharp aromatic odor.

Cinnamon of the Ceylon type is cultivated in Guyana, the Isle of St. Vincent, Cape de Verde, Brazil, the Isle of France, Pondicheny, Guadaloupe and elsewhere. There is, however, no probability that the tree will succeed as an article of commerce that has not the hot, damp insular climate and bright light of Ceylon.

The barks of all these different species, including that of Ceylon, are classed as “cinnamon” in the pharmacopias of Austria, Germany, Hungary, Russia, the United States, France, Spain, Denmark and Switzerland, while in the United Kingdom cinnamon must be the bark of the Ceylon plant *C. zeylanicum*; the others being classed as cassia.

Oil of Cinnamon

Oil of cinnamon is made from the pieces and chips of the bark, it is of a red-yellowish color. Eighty pounds of bark yields about 8 ozs. of oil. It is very stimulating. It is often adulterated with oil of cassia, oil of cassia buds, oil of cherry laurel, and oil of bitter almonds—the latter is a very dangerous mixture.

Cinnamon leaves yield an oil resembling oil of cloves, with which it is often mixed.

The ripe berries of the cinnamon tree yield a volatile oil, similar to oil of juniper, and from the root is obtained camphor.

Cassia oil is obtained from the leaves, buds, or bark. It is of a golden-yellow color, but turns brown with age. It is considered good for influenza.

Cassia buds resemble nails with heads of different size and shape, according to the period of growth when collected.

There is also a kind of wild cinnamon, or cassia, which grows in Cuba, but its taste resembles more that of cloves than of cinnamon.

CLOVES

Interesting History With Illustration Showing Flower, Bud and Fruit—Where Grown and Commercial Uses

Cloves are dried, unopened calyces or flower buds of the clove tree, *Caryophyllus aromaticus*, a kind of myrtle, a native of the Molucca islands. In commerce they are chiefly distinguished by their place of growth and rank in the following order: Penang, Bencoolen, Amboyna, and Zanzibar. In addition to these there enter into commerce as secondary products, clove stalks and mother cloves, or the dried ripened fruit. The bulk of these secondary products are shipped from Zanzibar.

The clove tree is an evergreen, 15 to 30 ft. high. It has a thin smooth bark and adheres closely to the wood, which is a gray color and of little use. The leaves are 3 to 5 in. long. The upper side and foot-stalk is red, shading to a dark color, while the under surface is green. The flowers grow in small bunches at the extremities of the boughs, very like the flower-buds of the lilac tree, and all are of a delicate purplish color. The calyx is long and forms the seed sack. As the blossoms fade the calyx changes color from yellow to red. If allowed to remain on the tree after this the calyx swells like that of the rose. In this state it loses its pungent properties and is called mother clove, and is practically of no value as a choice spice. The cultivated trees are kept pruned to about 8 or 10 ft. in height.



The harvesting of the flower-buds commences immediately after they assume a bright red color. Such blossoms as can be reached are plucked by hand, while those that grow on the upper branches are beaten down with bamboo poles and caught in clothes spread beneath the trees. They are then dried in the shade or by hanging on hurdles over slow wood fires—they lose about half their weight in the drying process. They are usually finished off in the sun, which gives them a darker color. The quicker they are dried the less the loss of aroma. Good cloves have a strong aromatic smell, a hot, spicy taste and a light brown or tan color. The season for harvesting is from September to March. A 10-year-old tree yields about 20 lbs. of cloves a year, the yield increasing up to 100 lbs. for a 20-year old tree.

Penang cloves are from the Straits Settlements. They are large, plump and of a bright color. Amboyna cloves are not so large as the Penang and are of a dark brown color. Zanzibar cloves are smaller than the Amboyna, a

bright reddish color and generally very dry. Pemba cloves are small and dark in color and mostly arrive in a damp condition, and therefore lose weight if kept long.

Cloves have sometimes a portion of their oil extracted, which gives them a pale, thin, shriveled appearance, altho they may be freshened up by rubbing with a little oil or passed off by mixing with good cloves. Cloves that have been tampered with have a good proportion of their heads or knobs off; altho another cause for headless cloves is that they may have been gathered when too ripe.

Pure oil of cloves is almost colorless, with a faint yellow tinge and the strong smell and burning taste of cloves. When old it turns to a reddish brown color. It has a greater specific gravity than water, in which it will sink.

Clove stalks and mother cloves are used in the manufacture of ground cloves and mixed spices. In Brazil the flower-buds of the tree whose bark furnishes cloves cassia are often used as substitutes for true cloves. The clove tree attracts so much moisture that herbage will not grow beneath its branches and the clove of commerce has such an affinity to water that if placed near a vessel of water they will absorb enuf of the moisture in a few hours to appreciably increase their weight. It is said that dealers often take advantage of this to increase the weight of their goods and thus enhance their profits.

A Little Clove History—This spice was well known to the ancients and is mentioned by several Chinese authors as in use under the Han dynasty, B. C. 266 to 220, during which period it was customary for the officers of the court to hold the spice in their mouth before addressing the sovereign, in order that their breath might have an agreeable odor. At this period the clove was called fowl's tongue spice. In 1265 A. D. the price was 12s per lb. In 1609 a ship of the East India Co., called the Consent, brot 112,000 lbs. to England which was sold at 5s 6d per lb. As was the case with nutmegs, the Dutch attempted to control the business in cloves. With this object in view, they caused all the clove trees to be destroyed except those of the island of Amboyna. The natives of the island were compelled to rear a certain number of plants each year and also to protect the bearing trees. The French, however, found a number of clove trees growing wild in

the smaller island, and Poivre, French governor of Mauritius, who obtained the plant from the island of Guebi, introduced the clove tree into that colony in 1770. About 1800 an Arab named Harameli-ben-Selah took some seeds and plants from Boubon to Zanzibar and commenced the cultivation of cloves in that country. The word clove is derived from the Latin clavus nail, Spanish clavo and French clou, owing its nail-like appearance.

GINGER

Used as a Spice by the Early Greeks and Romans—Plant a Native of Asia and Grew Wild in Mexico and Africa

As a spice, ginger was used among the early Greeks and Romans, who appear to have received it by way of the Red sea, inasmuch as they considered it to be a production of southern Arabia. In the list of imports from the Red sea into Alexandria which, in the 2nd century of our era, were then liable to the Roman fiscal duty, ginger occurs among other Indian spices. It appears in the tariff of duties levied at Acre in Palestine, about 1173, in that of Barcelona in 1221, Marseilles in 1228 and Paris 1296. It was known in England before the Norman conquest, being frequently named in the Anglo-Saxon leech-books of the 11th century as well as in the Welsh “Physicians of Myddvai.” During the 13th and 14th centuries, it was, next to pepper, the commonest of spices, costing on an average 1s 7d per lb., or about the price of a sheep. Three kinds of ginger were known to Italian merchants about the 14th century: (1) Belledi of Baladi, an Arabic name which applied to ginger would signify country, wild, and denotes common ginger; (2) Columbonio, which refers to Columbuno, Kolam or Quilon, a port in Travancore, frequently referred to in the middle ages; (3) Micchino, which denotes brought from or by way of Mecca. Marco Polo saw it in India and China, 1230–1239. John of Montecorvino, a missionary friar, who visited India in 1290, gives a description of the plant and refers to the root being dug up and transplanted. Nicolo de Conti, a Venetian merchant, early in the 15th century describes the plant and a collection of roots he saw in India. The Venetians received it by way of Egypt, and superior kinds

from India overland by the Black sea. Ginger was introduced into America by Francisco de Mondoca, who took it from the East Indies to New Spain. It was shipped for commercial purposes from the islands of St. Domingo in 1585, and from Barbadoes in 1654.



Ginger is the dried, knotty fibrous rhizomes or tubers—“races” or “hands” as they are called from their irregular, palmate form—of the ginger plant (*zinziber officinale*) the real roots being the thin fibers that branch off from the rhizomes.

The plant is a native of Asia, but also found growing wild in Mexico and East Africa. It is a reed-like biennial plant, not unlike the iris or flag in appearance. The leaves are long, similar to those of maize, growing alternate on a stem 3 to 4 ft. high. The flowers are borne on a separate stem, 6 to 12 in. high; they are yellow or blue, according to the quality of the soil

in which they have been grown. The plant which produces the yellow flower and best ginger is grown on rich, deep, virgin soil; the other comes from poorer ground. Ginger is propagated by pieces of the rhizome being planted in March. The flowers appear about September, after they have withered and seeded. The roots are dug up about January. When left too long in the ground, the rhizomes become very fibrous, if taken up too soon they are tender and succulent, so much so that they cannot be made sufficiently dry to render them fit for export in the usual commercial form. They are therefore preserved in sugar. The rhizomes, besides being classed as “yellow” or “blue,” are also divided into “plant,” (being the rhizomes from plants of the same season’s growth), and “ratoon” which are rhizomes left in the ground from the previous harvest.

Ginger is known in commerce in two distinct forms, termed respectively as coated or uncoated ginger,—as having or wanting the epidermis. For the coated ginger, the races of hands, after being dug up, are thoroly washed to free them from all the adhering earth. They are then laid on a canvas or cement floor, outdoors, to dry by the heat of the sun. At night they are taken indoors. It takes from 6 to 8 days to thoroly dry them. They are then ready for shipment. In damp weather they are artificially dried by an evaporator. In this form ginger presents a brown, more or less wrinkled or straited, surface, and when broken up shows a dark brownish fracture, hard, and sometimes horney and resinous. For the uncoated ginger the fresh-dug rhizomes, after being washed, are soaked in water for some time and then peeled or scraped—a most delicate operation requiring the hand of an expert. Owing to the peculiar formation of the races, no machine has yet been invented that will do the work satisfactorily. The outer rind or skin is deftly taken off by means of a common knife, so as not to injure the inner root, as a loss of the pungent volatile oil, to which ginger owes its value, would follow and thus impair its commercial worth. After being peeled the races are soaked in water over night. In the morning they are again washed, cleaned and weighed, and then dried in the same manner as coated ginger.

It requires 3 lbs. of green root to make 1 lb. of dry root. The purer the water the whiter the ginger. Sometimes lime juice is added to the wash water, which gives a whiter root, but as lime juice contains sugar, it prevents thoro drying and mildew follows. Ginger is often subjected to a system of bleaching, or by immersion for a short time in a solution of

chlorinated lime. The white-washed appearance which much of the ginger has is due to the fact of its being washed in whiting and water or even coated with sulphate of lime. Uncoated ginger varies from single joints an inch or less to flattish, irregularly branched pieces of several joints, the races of hands, and from 3 to 4 in. long. Each race has a depression on the summit showing the former attachment of a leafy stem. The color, when not white-washed, is a pale buff. It is somewhat rough, breaking with a short, mealy fracture, and presenting on the surface of the broken parts numerous short or bristly fibers.

The best ginger grown comes from Jamaica. It is of a superior strength, fine flavor and a light, handsome color. A peculiar trade custom prevails in Jamaica with regard to ginger, which is not sold by weight or measure but by the "heap," and the size of the heap governs the price and is an indication, to a certain extent, of the quality and quantity of the crop. If the heap is small, the price is high; if the heap is large, then the price is lower. If the races or hands, are finely shaped and large, there are fewer in the heap; if small, dark and mealy, the heap is made larger.

The next best quality is Borneo or Cochin ginger, which closely resembles in appearance the Jamaica. It is not, however, so carefully prepared.

African ginger, also termed Bombay or Calcutta, from the ports of shipment, is darker in color, has a coarser appearance, a harsher flavor and inferior aroma to either of the above, but contains a greater amount of oleoresin than they do and is very pungent. It is largely used for making ginger beer, essences, extracts, etc.

Leaf ginger is ginger that has been sliced into thin flakes.

Green ginger root, is that which has not undergone any process of cleaning beyond freeing it from the earth adhering. Imported in casks and used by wine makers, preservers, etc.

Spent-ginger is whole ginger that has once been used, then fixed up to resemble good ginger and sold whole or ground. It does not possess a single one of the valuable properties of genuine ginger.

China ginger is not imported in a dried state, the rhizomes being too tender and succulent to thoroly dry for export. It is preserved or candied.

For preserving, the rhizomes are first scalded, then washed in cold water and peeled, then boiled in pans for 2 or 3 hours; then transferred to copper pans and boiled for 2 hours in a mixture of sugar and water—just sufficient water to cover the roots, 5 lbs. of sugar to 10 lbs. of ginger, the roots having been pierced with a sharp instrument to enable the sugar to soak into them. After boiling the ginger is put into large jars and stands for several days, when it is again boiled in sugar and water in the same quantities. After it has become cold it is packed in jars or tins for export. To crystallize, the same process is gone thru, only in the final boiling it is boiled until the sugar become dry.

The Chinese season for preserving ginger is from July to October. It is nearly all prepared in Canton and Hongkong. A kind known as Ng Mai Keunig is preserved in Swaton, from *Alpina galanga*, but it is not like the Canton or Hankou ginger and is only made for native consumption, to be used medicinally or for cooking. Some of it goes to the Straits Settlement, but none to Hongkong. Jamaica preserved ginger is mostly put up in glass bottles. The uses of ginger are too well known to need repeating.

MUSTARD

Well Known to the Ancients, but More in a Medicinal Way—How Cultivated and Prepared for Commercial Uses

Mustard was well known to the ancients, but more in a medicinal way than dietetic. From an edict of Diocletian, 30 A. D., in which it is mentioned along with alimentary substances, we must suppose it was then regarded as a condiment, at least in the eastern parts of the Roman empire. In Europe, during the middle ages, mustard was a valued accompaniment to food, especially with the salted meats which constituted a large portion of the diet of our ancestors during the winter. In the Welsh “Meddygon Myddrai” of the 13th century, a paragraph is devoted to the “Virtues of Mustard.” In household accounts of the 13th and 14th centuries, mustard is of constant occurrence; it was then cultivated in England, but not extensively. The price of the seed between 1285 and 1340 varied from 1s 3d to 6s 8d per quarter (21 lbs.), but between 1347 and 1376 it was as high as 15s and 16s. In the accounts of the Abbey of St. Germain des Pres in Paris, 800 A. D., mustard is specially mentioned as a regular part of the revenue of the convent lands.

The essential oil of mustard was first noticed in 1660 by Nicolas le Febre and more distinctly in 1732 by Boerharroe.

The word mustard comes from the Italian, murtard, which is derived from the Latin must-um, unfermented grape juice, with which the Italians formerly mixed ground mustard. The Athenians called it napy; while the Hellenistic name was sinapi, or sinapy, whence the Latin sinapi, or sinapis, from which is derived the German word senf. Hippocrates used mustard in

medicine under the name of Vanuit. The dark seed, which comes from Trieste, Austria, is called Trieste mustard. Spoken of by Theophrastus, Galin and others. What is called French mustard, German mustard, etc., is made of the dressings mixed with vinegar, garlic and other spices and flavoring material. The form in which table mustard is now sold dates from 1720, about which time Mrs. Clements, of Durham, Eng., hit on the idea of grinding the seed in a mill and sifting the flour from the husk. This bright yellow farina rapidly attained wide popularity. The fame of "Durham Mustard" was spread far and wide, Mrs. Clements traveling to London and principal cities twice a year taking orders.

There are two species of mustard plants from which ground mustard is made. The *sinapis alba*, white or yellow mustard, and *sinapis nigra*, brown or black mustard, is the mustard plant spoken of in Luke XIII, 19. They are annual herbs, three to 6 ft. high, with lyrate leaves, yellow flowers, and slender pods, from one to four inches long, containing a single row of roundish seeds.

One of the peculiarities incident to the cultivation of mustard is the fact that two crops of mustard cannot be raised on the same ground in succession. Another variety is *sinapis arvensis*, or wild mustard, called charlock and used for adulterating; the *Sarepta*, the black seed of the *sinapis juncea*, from the East Indies, is used for the same purpose. *Sarepta* is called from a city of that name in Russia, in the government of Saratov.

The brown or black variety is sown in January and the yellow or white in March, the seed being sown broadcast and harvested in August. A reaper is used, cutting the stalks and throwing them in bunches, where they are left to cure until October. They are now thoroly dry and are taken to a convenient place, spread out upon sheets of canvas and rolled with a heavy roller. The stalks and empty pods are then raked off, and the chaff and seeds remaining are run thru a fanning machine, after which process they are ready to sack and market.

There are two processes in use in making ground mustard. In the first, the seeds, white or black, or mixed, are ground to powder and then put thru an elaborate course of siftings. The product left after the first sifting is called "dressings" and that which passes thru is pure mustard flour. This mustard flour is again run thru a finer sieve, and so on until the required

fineness is obtained. From the dressings left after the different sievings, the essential oil of mustard is expressed.

In the other method, the oil is first extracted from the seeds by hydraulic pressure, which leaves a sort of cake. This cake is then broken up and pounded in a mortar. It is then sifted, that going thru the sieve being a kind of bolted mustard flour. The remaining bran is then mixed with an equal quantity of wheat flour, one per cent of cayenne and sufficient turmeric to give the proper color. This is pounded and treated as before, the process being continued until there is no bran left. Then all the different siftings are mixed together, giving a mixture of about equal proportions of mustard and wheat flour, with the cayenne and turmeric added in proper quantities.

The peculiar pungency and odor, to which mustard owes much of its value, are due to an essential oil developed by the action of water on two chemical substances contained in black mustard seed; one called sinigrin and the other myrosin. The latter substance in the presence of water acts as a sort of ferment on the sinigrin, and it is worthy of remark that this reaction does not take place in the presence of boiling water and, therefore, it is not proper to use very hot water in the preparation of mustard, cold water only should be used. White mustard seed contains in the place of sinigrin a peculiar acrid substance called sinalbin and also a trace of myrosin, therefore, it possesses very little pungency and it produces a larger percentage of flour than the black. The proper blending of these two seeds is necessary to the production of the best mustard, as the white has the peculiar ferment within it which develops to the highest degree the flavor of the black.

The reason for mixing wheat flour, rice flour or other farina with pure mustard flour is, that owing to the large amount of oil contained in the latter it will not keep long, but turns rancid, ferments and cakes; the added farinas by absorbing a portion of the oil retards fermentation, decomposition and rancidity. They should not be looked upon as adulterants, unless added in too great quantities, and the price of the mustard should be in proportion to the added absorbents.

A mean form of adulteration is to mix gypsum and chrome yellow with the ground mustard seed.

If upon the addition of a small quantity of iodine to ground mustard it turns blue, it shows that starch is present. The ammonia test will show the presence of turmeric. Every manufacturer has his own particular formula, and consequently there are many different qualities, both in the pure mustard and the compounds. One is composed of 37 per cent brown and 50 per cent white mustard flour, 10 per cent of rice flour, 3 per cent of black pepper, a little Chili pepper and ginger.

Pure mustard oil, as pressed from the seed, is not pungent and will not blister unless mixed with water.

The English mustard seed is the best in the world. Of this class 4,995,800 lbs. of seed and 1,307,202 lbs. of flour were imported during the year 1908. Mustard seed and flour from Italy is known as Trieste. In the Lompoe valley, California, some 2,500 acres are under mustard cultivation, and a small quantity is also grown in Kentucky.

The uses of mustard are too well known to need recapitulation. D. S. F. means double superfine.

NUTMEG AND MACE

Where the Nutmeg Tree Grows—Yield of Nuts and Mace and How Prepared for the Market—Uses in Commerce

The nutmeg tree, known to botanists as *Myristica fragrans* (sweet smelling) is a native of the Malay archipelago. The tree, which in the Banda isles grows to the height of 50 to 60 ft., and in the Straits to 30 to 40 ft., resembles the pear tree in the shape of its leaves and fruit. Its flowers are like those of the lily of the valley in form and size, but are pale yellow and exceedingly fragrant. There are male and female flowers, the nutmegs being obtained from the latter. It is only when the tree is about 6 or 8 years old that the female tree can be distinguished from the male, and of the latter only a few are allowed to remain for fertilizing purposes, the rest being cut down. The nutmeg tree continues to yield from 70 to 80 years after reaching maturity (8 years). Each tree on an average will produce 10 lbs. of nutmegs and 1½ lbs. of mace annually. The fruit is yellowish, edible drupe, about the size of a peach; it splits into halves when at maturity—about 9 months from the time of blossoming—exposing a single seed with a thin, hard shell, surrounded by a fibrous substance of a crimson color, which, when dried and shelled becomes the nutmeg. The young drupes, when young and tender, are often preserved like jam and are considered the most aromatic and delicious of conserves. Altho the nutmeg tree has ripe fruit upon it at all seasons, there are three principal periods of harvesting, viz: July, when the fruit is most abundant, though it yields thin mace; November, when the mace is thicker, though the nutmegs are smaller, and March, when both

mace and nutmegs reach their greatest perfection—but as this season is dry the production is not great.



The usual method of gathering in the Straits is to collect the ripe nuts that have fallen on the ground. In the Banda islands, the fruits are gathered in small, neatly-made, oval bamboo baskets—holding about 3 fruit—at the end of a long bamboo stick, which prevents bruising, the baskets being opened for about half their length on one side, and furnished with two small prongs projecting from the top, by which the fruit stalk is broken, the fruit falling into the basket. After the pulp—which is about $\frac{1}{2}$ -in. thick, whitish in color, and tough like candied peel—has been removed the mace is stripped off by hand. The shell of the fruit is very hard and cannot be broken without injury to the kernel. To overcome this they are put into receptacles with fine mesh bottoms, and dried over a slow fire—being turned from time to time—until the kernel rattles freely in the shell, a process which takes about 6 or 8 weeks. This also kills any weevil which may be at work in them. They are then carefully cracked by placing them on a sort of drumhead made of raw-hide and striking them with a board or mallet, when the shells fly off into pieces. Great caution is needed in shelling, for if too hard a blow be struck it makes a black spot on the nutmeg, which affects its value considerably. After being steeped in salt water several times and again dried they are sorted according to size and soundness—130 to 140 to the pound are the lowest priced, 75 to 80 the highest, and larger nuts are sold at special prices. The sorting is done by hand, and nothing but sound, perfect nuts are supposed to be shipped. The broken and wormy ones are used in manufacturing “nutmeg butter,” or, as it is commonly but erroneously called, “mace oil.” They are now limed. There are two methods of liming in vogue—the dry and the wet. In the dry process, the nuts have dry lime powder rubbed over them, either by hand or shaking in barrels. In the wet process, the nuts are put into newly-slacked lime and then spread out to dry, or they are dipped into a kind of lime-pickle, thick as syrup, made of calcined-shells and salt water. After being covered with this mixture they are dried. The process of liming originated

with the Dutch, with a view to preventing the germinating of the seeds, for which purpose they were formerly immersed for three months in milk of lime. Again it is claimed that liming preserves the nuts against the attacks of maggots and a particular kind of beetle by stopping up their breathing and chewing apparatus. A preference is still manifested for limed nutmegs.

As nutmegs are now seldom shipped by sailing vessels, but by steamers, thus saving the long-time voyage, there is no reason why they should not come unlimed, and then the differences in their natural complexions and range of variations would become familiar and easily recognized. The liming process hides many imperfect or corky nuts; nuts which have been riddled with worm holes are “stopped” with a paste made of flour, oil and nutmeg powder and then mixed with the sound ones. Occasionally this paste is moulded into false nutmegs. Besides this, nutmegs are frequently robbed of part of their essential oil by distillation in alcohol—a process called “sweating”—and yet sold as entire nuts. A small quantity of boracic acid will accomplish the same purpose as lime, and Paris white and barytes will serve to mask the identity as well as the defects. A good nutmeg should have no worm-holes, be full of oil and cut firm like a piece of wood, and if a pin is thrust into one the oil should ooze out on its being withdrawn.

The Penang nutmegs, which are generally not limed, are considered the best, altho some prefer the Banda or Batavia, and after these the Singapore. There is also a demand for an elliptical-shaped nutmeg of rank flavor, first called long nutmegs, but now known as Macassars. Another kind of nutmeg from New Guinea, and known in Germany as “horse nutmeg,” is from the species *Myristica Argentea*. It is of a long and narrow shape. In these the arellus or mace furrows are less marked and their odor is not so delicate as that of the true nutmeg.

There are many kinds of wild or inferior nutmegs, such as: American Jamaica, or calabash nutmeg (*M. monodora*), of the custard-apple family, bearing a large pulpy fruit containing aromatic seeds. Brazilian nutmeg (*cryptocarya moschata*) a tree of the laurel family, producing nutmegs of an inferior quality. The nut is longer than the true species and is sold under the name of long nutmeg. California nutmeg, a tree of the pine family, called also stinking nutmeg or stinking yew, from the disagreeable odor of the leaves and wood when bruised and burned, and yielding a fruit resembling

true nutmegs. Clove nutmeg, a Madagascar tree of the laurel family, the fruit a pungent kernel resembling the true nutmeg and used as a spice. Peruvian nutmeg, a large tree of the monimiad family, yielding an aromatic fruit. From Borneo a wild, soapy nutmeg and mace (*M. fatua*) are often palmed off as the true kinds. There is also the Sante Fe nutmeg (*Motoba*) from Columbia, S. A., and Ackaway nutmeg, a spice grown in Guiana, the fruit of *Acrodictidum camard*. Another species, the *M. sebefira*, is a common tree in the forests of Guiana, north Brazil, and up into Panama. It is utilized principally for the oil extracted from the nuts, obtained by macerating them in water, the oil rising to the surface, and as it cools skimmed off. Ackawi nutmegs, used mainly as a cure for diarrhoea and colic. All these, while resembling somewhat the true nutmegs and sometimes foisted on dealers, are of very little real value.

Mace

When the mace, a bright-red membraneous substance, is removed from the nut it is pressed flat between blocks of wood and left to dry until it has acquired the right color. The preparation of mace for the market requires experience rather than technical knowledge. If packed too green it is liable to mold, and is subject to attacks from insects, which render it valueless in commerce. On the other hand, if it becomes too dry it loses its vitality and also crumbles into powder when packed. Packers frequently sprinkle the mace with salt water, which makes it more pliable and at the same time prevents attacks from insects.



We may here state that nutmegs are divided into two varieties: The green, which are long and in which the mace only partially covers the nut; is darker in color and inferior in flavor and aroma; and the Royal, which furnishes the finest and best mace, firm, thick, flexible and oily, and entirely envelopes the nut.

As with the nutmeg, mace is sometimes deprived of its essential oil, and mixed with wild mace or other flavorless matter. *Myristica Malabarica*, known under the name of Bombay mace, used to adulterate the true powdered mace, is much larger and more cylindrical than the arillus of the

true nutmeg and has several flaps united at the apex, forming a conical structure.

Products—Candied nutmeg and mace, nutmeg fruits in vinegar or salt, preserved nutmeg fruits, and nutmeg or mace essence made from the essential oil of nutmegs (not mace) and rectified spirits. An essence of mace is also made from 6 oz. mace and 2 pints cologne spirit, macerated for a couple of weeks, expressed and filtered thru paper. “Nutmeg butter”, “butter of nutmeg,” or mace, “concrete oil of nutmeg,” or “expressed oil of mace,” as it is variously called, is obtained by subjecting the nutmeg or mace to a great heat and then squeezing or pressing it in heavy presses. This substance is of a green color of the consistency of tallow and of a pleasant smell. A pound of nutmegs will make 3 ozs. of this oil, but a transparent volatile oil is obtained by distillation. It evaporates rapidly on exposure to air. When cold it becomes somewhat spongy and has a marbled or mottled appearance. It becomes hard with age and is exported in small bricks, 10 in. by 2½ in., wrapped in palm leaves. It is known under several names, as nutmeg butter, balsam of nutmeg, concrete oil or the mace oil of commerce, and as Banda soap, sometimes made from the distilled nutmeg leaves, counterfeited by using a foreign fatty substance as palm oil, nut, wax and animal fat, boiled with powdered nutmeg and flavored with saffron, which gives it the right color and flavor.

Uses—Nutmegs, besides their use as a spice or condiment, are used to relieve sleeplessness when opium fails and chloral is not advisable. For diarrhoea, half a drachm in milk is an effective cure. Butter of mace is used as a liniment and embrocation for rheumatism and is also a favorite medicine for low stages of fever with Hindoo doctors.

For ground nutmeg, all the faulty, broken, moldy, worm-eaten and wild nutmegs are often used.

A little of the history of mace and nutmegs: It has generally been believed that neither the nutmeg or mace were known to the ancients. Nutmegs and mace were imported from India at an early date by the Arabians, and thus passed into western countries. Masudi, who appears to have visited England in 916–920 A. D., pointed out that the nutmeg, like cloves, arca nut and sandalwood, was a product of the eastern isles of the Indian archipelago. The Arabian geographer, Edrisi, who wrote in the

middle of the 12th century, mentions both nutmeg and mace as articles of import into Aden. They are also among the articles on which duty was levied at Acre in 1180. About a century later another Arabian author, Kozwim, expressly named the Moluccas as the native country of the spices under notice. One of the earliest references to them in Europe occurs in a poem about 1195, by Petrus D'Ebulo, describing the entry into Rome of the Emperor Henry VI, previous to his coronation in 1191. By the end of the 12th century both nutmeg and mace were found in northern Europe, even in Denmark, as may be inferred from the allusions to them in the writings of Harpestring. In England, mace, though well known, was a very costly article, its value between 1284 and 1377 being about 4s 7d per lb., while the average price of a sheep during the same period was about 1s 5d, and of a cow 9s 5d. It was also dear in France, for in the will of Jeanne d'Evreux, queen of France, in 1372, 6 ozs. of mace were appraised at the rate of 8s 3d per lb. In the middle of the 18th century, the Dutch, with the object of monopolizing the trade in nutmegs, destroyed all the trees in all the Moluccas islands, excepting Banda. Nature did not, however, sympathize with such meanness. The nutmeg pigeon, found in all the Indian islands, did for the world what the Dutch had determined should not be done—carried the nuts, which are their food, into all the surrounding countries, and trees grew again and the world had the benefit. In order to keep up the price, the surplus stock was burned up each year by certain unscrupulous men, as is proposed to do at the present day with the surplus stock of Brazilian coffee. In 1760, they burned at Amsterdam three such immense piles of nutmegs and cloves that one writer says: "Each of which was as big as a church."

This account of nutmeg would not be complete without "Connecticut Nutmegs." Some 90 years ago Frederick Accum startled England with his book "Adulteration of Food and Culinary Poison," and a sort of pure food hysteria passed thru the country similar to that caused by the boric acid investigation here. But he was eclipsed by a person who declared that the makers of wooden shoe-pegs in Connecticut were making oats and nutmegs from the discarded wood of sawmills. He asserted they were not only made, but used as food thruout the country. Thus was Connecticut christened the Nutmeg State, a name which it has retained even unto this day.

PEPPER

White and Black Varieties and Why— How the Plant Is Cultivated and Where —History the Grocer Should Know to Judge Qualities

Pepper is a commodity to be found in every grocery store, but how many grocers know that the pepper plant—*Piper nigrum*—which produces the white and black pepper of commerce, is a climbing vine-like shrub, found growing wild in the forests of Travancore and Malabar coast of India? It is extensively cultivated in southwest India, whence it has been introduced into Java, Borneo, the Malay peninsula, Siam, the Philippines and the West Indies.

Pepper in the time of Alexander the Great was considered an extremely choice article and, like gold and precious stones, was for many generations found only on royal tables. During the Middle Ages, it was used as money in payment of tolls, etc., hence the custom of “pepper corn” rentals, i. e., a nominal rental or perpetual lease; and its high price is said to have been one of the causes which led the Portuguese to seek a sea passage to India.

The pepper plant grows naturally to 20 ft. in height, but is cultivated on trellises or poles, about 10 or 12 ft. high and is propagated by cuttings or suckers. It has a soft stem, the leaves are 4 to 6 in. long, tough, glossy, broadly ovate, with 5 to 7 nerves, and grow opposite and alternate to a pendulous spike 5 to 8 in. long, having 20 to 50 white flowers that ripen into a one-seeded fruit with a fleshy exterior. This fleshy berry, covering a soft stone, is about the size of a pea and is at first green, but in ripening

turns red, which gradually darkens to a deep chocolate shade. The vine begins to bear when 3 or 4 years old and continues bearing for the next 10 or 15 years. It is in perfection at its eighth year.



There are two crops a year—July and December—which yield 5 to 6 lbs. of dried pepper each for a single vine. When the berries are ripe the stalk is pinched off by hand and placed in an oblong cane basket, slung horizontally behind the plucker by a rope around his waist. The rounded ends of the basket extend a little on either side, so that the basket can be easily filled by either hand of the workman. The berries are rubbed off the spikes by hand and placed on mats or on the bare ground, to dry in the sun,

when the weather is fair. In damp or cloudy weather they are placed in shallow, open baskets before a gentle fire. If the berries are left too long on the vines they lose part of their aromatic, pungent hot taste, and if gathered too soon they become broken and dusty in drying. After drying, when they become black and shriveled up, they are cleaned and winnowed. Good black pepper is firm and not too deeply wrinkled, does not easily crumble or break in the hand, it is also heavy and readily sinks in water. The inner seed should be hard, round and smooth and of a grayish-brown color. The outside pericarp should be brownish-black. A yellow tinge betrays over-ripeness and consequent loss of strength. A reprehensible practice among some dealers to hide defective peppers is to artificially blacken them and polish with oil. The usual method of judging quality is by weight, the grades technically being known as heavy, or shot, half-heavy and light peppers or corns. A one-litre measure may be filled with the pepper and the contents weighed, or 100 corns of average size counted and their weight ascertained. The variations of peppers of different qualities, according to their habitat, are given in the following table:

Variety—	Weight per litre
Singapore	476 grams
Tellicherry	548 ”
Lampong	511 ”
Mangalore	574 ”
Malabar	570 ”
Acheen	407 ”

It is evident that the moisture present in the corns plays an important part in the determination of the weight, and it will be necessary to bring the peppers up to the stated water content by either drying them or placing them in a moist atmosphere, or first weigh them dry and weigh again. A slight variation, however, from the figures given, is unavoidable.

Singapore Pepper—The principal part of this import is the product of Sumatra, Borneo and Siam, collected at Singapore. A considerable quantity, however, is the products of the Straits Settlements themselves. It is of large

size and of a fairly uniform quality, but as pepper powder it is not much esteemed, owing to the manner of drying, giving it a smoky flavor that buyers can distinguish Singapore pepper from peppers grown elsewhere.

Tellicherry and Alleppey are much alike in appearance, both being light brown in color. They too, like the Malabar peppers, are sun-dried. Mangalore (India) pepper is heavy, large, of a deep black color, very clean, and of uniform size. When powdered it is of a greenish-black appearance.

The pepper shipped from Penang is called Irang pepper and is grown in Sumatra. From the east end of the same island comes the Lampong pepper, but this lacks uniformity, and is light in color. It is also sun-dried. Long pepper is the fruit spike of *Chivaci* Roxburgh, a native of Malabar and *Chavica officinarum*, a native of the India archipelago; they are both climbing plants. The first pods, or catkins, about 1½ in. long, grow nearly straight, and opposite the leaves. They are gathered before they are ripe and dried in the sun, when they become brown or dark green in color and rough to the touch. They lack the pungency of the black variety. The long pepper plant dies at the end of 3 years, and after the fruit is collected the vine dies down to the ground. The fruit grows so close together on the spike that when ripe they become one solid mass. There is also a variety of long pepper called elephant pepper. Long peppers are mostly used for pickles. A medium, called Pippua moola, is made from the roots and stems; it is very stimulating.

Cubeb peppers are the berries of the vine *Cubeb officinalis*, a product of Java, Borneo and Sumatra, but mostly imported by way of Batavia and Canton. They are of a gray color, about the size of black pepper, somewhat longer, more wrinkled and with a short slender stalk. They have a hot, camphor taste. Another kind is distinguished by a mace-like odor and taste. Cubebs are now mostly used as a medicine.

Ashantee or West African pepper is the dried berry of a pepper plant which grows in tropical Africa. It is smoother and smaller than the black pepper and resembles the Cubeb very closely. In taste it resembles the ordinary black pepper. At one time its importation was forbidden by the king of Portugal, as it threatened to interfere with the commerce of India.

Betel pepper is the berry of *Chavica betel*, a species of climbing vine largely cultivated in the East Indies, Ceylon, Burma, Siam, etc. It furnishes

the leaves which are used along with arecanut and other ingredients to compose the favorite stimulant chewing mixture of the people of India.

White pepper is from the same plant as black pepper, with the difference, that to make white pepper the pepper corns are not picked until fully ripe; they are then soaked in water for 7 or 8 days, or heaped up so that the pulp ferments, then they are rubbed by hand, or on a coarse cloth, if the quantity be small, or trampled under foot if the quantity be large; this operation deprives them of the pulpy skin or husk, and the greenish-white seeds which remain are the white peppers of commerce; then they are re-dried, either in the sun or by artificial heat. White pepper is bleached whiter by a chemical process. If the berries be left on the vines until over-ripe they lose their pulpy husk by natural decay and thus become actually white pepper, altho in reality they are the kernels of black pepper.

Singapore white are berries cultivated in the neighboring islands and the husks are removed at Singapore by hand and friction before the berries are fully dried. Penang white is really grown at Sumatra, but imported into Penang in a dried state. There the berries are soaked in lime and water for several weeks, until the pulp is soft, when it is rubbed off by hand and washing; the berries are then re-dried.

Siam white are berries prepared in the same manner as Singapore white, from berries grown in Siam.

The dried black peppers, as imported, are also decorticated or deprived of their husks by machinery, the result being white pepper, which is sometimes bleached.

The active properties of pepper are an acrid resin, a volatile oil, and a crystallizable, colorless substance called pipertine, or peperic. Why white pepper should be preferred before the black is one of the anomalies of the trade. White pepper has really only about a quarter the strength of black pepper, and is the least economical to use for these reasons: (1) Because of being allowed to ripen it loses much of its pungency. (2) Because it is deprived of the outer skin or husk, which contains much of the constituents which go to make good pepper. (3) Because it contains scarcely a trace of piperin, one of the most active principles of pepper. Pepper rapidly deteriorates under atmospheric influences, and large stocks should not be carried unless provisions are made for storing it in air-tight receptacles, for,

unless this precaution is taken, the goods in a few months will have lost their pungency, which is an essential characteristic of good pepper.

Pepper is a stimulant, and used in moderate quantities is an aid to digestion. In India an infusion of it is used to create an appetite and as a cure for gout and palsy. It is also used in cases of cholera-morbus. A liniment is made from the berries for rheumatism, and the root is employed as a tonic stimulant and cordial.

CUMIN, OR CUMMIN SEED

**Also Caraway, Coriander, Cardimons,
Poppy, Aniseed, Saffron and Turmeric
Described.**

Cumin, or Cummin Seed

The aromatic fruit or seed of a plant of the genus Umbellefera. It is referred to in Scripture (Matt. xxxiii:23). As salt was a symbol of friendship, “shearers of salt and cummin” meant intimate friends. The seeds are linear and flat on one side and convex or striated on the other. Their odor and properties resemble the caraway, or anise seeds, and they are often called bastard anise. They are used in Germany in bread, in Holland they are frequently put into cheese. Norwegian anchovies in kegs are frequently flavored with them, and they are also used in making curry powder, as a carminative flavoring, and in veterinary medicines, etc.

Caraway Seed

The caraway plant has a branching stem 2 or 3 ft. high, with finely divided leaves and dense umbels of white or pinkish white flowers. The leaves are frequently used to flavor soup and the roots, which taper like a parsnip, and when young are boiled and eaten as a vegetable. The seeds are oblong, pointed at both ends, thickest in the middle, striated on the surface

and of a crescent shape, they have an aromatic smell and warm, pungent taste. From the seeds is obtained a volatile oil called oil of caraway, of a pale yellow color which turns dark with age; it is frequently adulterated with oil of cumin. After the oil has been extracted the seeds are called “drawn caraways,” and by way of deception are often mixed with good caraway seeds. They can be told by their shrunken, dark appearance. The color of the English caraway seeds is a deep brown, those of Germany and Holland are larger and of a light blue-brown color, while those from Russia, Poland and Bohemia are small, of a blackish brown color, and mixed with a good deal of dirt. There is a variety of a light brown color, about twice the size of the English caraways, imported from Mogador.

Caraway seeds and oil are used medicinally, as a flavoring by bakers and confectioners, in compounding various liquors, particularly that known as Kummel, and in making Scotch cavie, or caraway, comfits; for this purpose the seeds are coated with sugar and colored red, pink, blue, yellow, etc.

Coriander.

The word “coriander” is derived from the Greek word Koriannon, a bed-bug, referring to the disagreeable smell of the whole plant when fresh, but the ripe and perfectly dried fruit has an agreeable smell and a sweetish, aromatic taste. It is an annual or bi-annual plant, of the genus Umbelliferce, native of South Europe, with a branching stem 1 or 2 ft. high. The lower leaves bipennate, the upper ones being more compounded and divided into very narrow divisions. The fruit is globose, containing round slightly ribbed or ridged seeds, about as large as black pepper, very light, of a yellowish brown or straw color externally; inside the husk of each seed are two closely fitting hemispherical mericarps.

The seeds are used in medicine as a carminative. They cover the taste of senna leaves better than any other substance; are occasionally mixed with curry powder; in domestic economy they are used by confectioners and bakers as flavorings, being often mixed with bread in the north of Europe. A

cordial is made from them, and they are used for flavoring spirituous liquors, particularly gin.

Cardamons.

Cardamons consist of the seeds of two species of plants, the *Elettaria* of Malabar and the *Amomon* of China, Guinea and other parts of the East Indies. As the seeds of the two species differ in some respects we will describe the *Ellettaria* kind. The plant, which grows 5 to 10 ft. high, has a reed-like habit and bear long, loose racemes of flowers, succeeded by triangular capsules, of a dirty white color, containing a number of dark brown, angular seeds about the size of mustard seeds. The capsules or fruits, which vary from $\frac{1}{2}$ in. to 2 in. in length, are collected from wild plants and also from plantations, the latter being generally laid out in partially cleared forests in which the wild plants are known to occur. When about 3 years old the plants begin to bear. The capsules do not all ripen at the same time, and the harvest lasts for nearly two months. The capsules are gathered before they are ripe and then cured in the sun, after which the stalks and remains of flowers are carefully removed by means of scissors. They are then graded into "shorts," "short-longs," and "long-longs," according to their length; sometimes they are mixed and classed as lesser or greater cardamons. Cardamon seeds are exported in the capsules in order to prevent adulteration. The seeds have a very delicate aroma and are slightly pungent. They were well known to the ancients, and are used at present in medicine, particularly in veterinary practice, also in flavoring culinary sauces, soups, curries, cordials, pastry, and for imparting a factitious strength to vinegar, beer, wines and spirits, especially gin; their use creates a thirst. The seeds depend for their quality on a pungent essential oil, of which they contain about 3 per cent, called oil of cardamons; they also contain about 10 per cent of a fixed oil. The seeds of the "*Amomum*" species of cardamons are bright black in color outside, white inside and small and angular in shape; they are slightly aromatic, very hot and pungent.

Cardamons are known as grains of Paradise, Melegueta pepper, Guinea grains and Guinea pepper.

Poppy Seeds

Poppy seeds are not unlike fine gunpowder in general appearance, being very small, dark blue—nearly black in color; they are obtained from the same plant that yields opium (*Papavar somniferum*, or white poppy.) The seeds are not narcotic, and have a sweet taste, are oleaginous and nutritious. They are largely used in some parts of Europe in pastry, confectionery and as a substitute for almonds. Under the name of “Maw seeds,” they are sold as food for birds during moulting season. Poppy seed oil is sometimes used as an adulterant in olive oil; it is also used as an illuminant and for painting.

Fennel

Fennel is a tall, stout, aromatic herb of the parsley family, with finely dissected leaves, which are boiled and served with salmon, mackerel, etc., as a seasoning; the flowers are yellow. A species—*F. dulce*—is cultivated in Italy as celery is with us; and its blanched stems are said to be more tender and delicate than celery, with a slight flavor of fennel. The seeds of another species—*F. panmorium*—grown in Bengal, have a warmish, very sweet taste and aromatic smell, and are used in making betel, in curries, and also used as a carminative. Fennel seeds resemble aniseeds in appearance and taste, and are often sold for such; they are a little longer and of a light brown color. The Indian seeds are the largest, the Italian and Japanese the smallest. They are used in confectionery, cookery and are sometimes chewed by the people of France and Germany. Fennel water is made from the oil obtained from the seeds.

And he who battled and subdued
A wreath of fennel wore.—Longfellow.

Aniseed

Aniseed is an annual plant of the order of Umbelliferae of the parsley family, a native of Egypt, but also extensively cultivated in Russia, Germany, Malta and Spain. Aniseed is very similar in appearance to the poisonous hemlock seed, for which it has sometimes been mistaken. The seed, which is a little larger than a pin's head, is of a greyish-green color. They have an aromatic smell, and warm, sweetish taste, and are used in condiments, in cookery and in the preparation of liquors, also in medicine as a stimulative stomachic to relieve flatulence, etc., particularly in infants. The properties of aniseed are due to a nearly colorless or sometimes blue volatile oil. Aniseed oil with water and sugar is much used in Italy as a cooling drink. The leaves of the plant are sometimes used as a seasoning and for garnishing.

Star aniseed, or China aniseed, is the fruit of a small evergreen tree of the order Magnoliaceae, somewhat resembling a laurel. It receives its name from the star-like form of the fruit or capsule, which consists of a number (6 to 12) of hard, woody, one-sided follicles or carpels ending in a point, each containing a single brown, shiny seed. Star aniseed is held in high esteem by the Japanese and is planted near their temples, the seeds being burned as incense in the temples and over the graves of relatives. The whole plant is carminative, and is used by the Chinese as a stomachic and as a spice in their cookery. The qualities of the seed and oil closely resemble those of the common aniseed and the oil is exported to Europe for the same purpose—flavoring liquors.

Saffron

Consists of the dried stigmas of the autumn or fall crocus plant (*crocus sativus*), which should not be confounded with the spring crocus (*crocus vernus*), to which it is nearly allied. The crocus derives its name from *Croceus*—which is from the Greek word *Krokus*, yellow—the modern *Korghi* in Cilicene, where it was grown in ancient times. The word “crocodile” is derived from the Greek words *Krokos*, yellow, and *deilos*, fearful, on the ancient supposition the animal avoided the place where saffron grows and only sheds real tears when in the vicinity of a crocus field, hence Fuller says: “The crocodile tears are never true, save he is

forced where saffron groweth.” The phrase, “crocodile tears,” arose from the idea that the crocodile pretended to cry over the victims it had devoured. Saffron was of great importance ages ago. It is mentioned in the third chapter of Solomon’s Songs; it was in favor among the ancient Greeks as a dye, and with both them and the Romans as a perfume. The streets of Rome were sprinkled with saffron when Nero made his entry into that city. In the middle ages it was employed in cookery and as a drug, and it is on record that as late as the fifteenth century persons were burned alive in Muremburg for adulterating saffron. It was introduced to England in 1339 from Tripoli by a pilgrim who had a stolen bulb in the hollow of his staff. Its main use was to color pastry and confectionery, hence: “I must have saffron to color the warden pies” (Shakespeare’s *Winter’s Tale*, act 4, scene 1). The town of Saffron Waldron in Essex, derives its name from the fact of its being cultivated in that neighborhood until 1768. The cultivation of the crocus for saffron in England has entirely died out; altho the people of Cornwall at the present day use more saffron than all the rest of Great Britain. It is cultivated in China, Cashmere, Persia, Asia Minor, Egypt, Austria, Hungary, Russia, Italy, France, but the chief source of supply is Spain.

A saffron field is not in full bearing until the end of the second year, at the end of the third year it is exhausted, and it is said that the soil is so poisoned that it cannot be used for any other crops for several years. Each acre produces from 600,000 to 700,000 bulbs and each bulb 2 or 3 flowers. About 150,000 flowers are required to produce 2 lbs. of fresh pistils, which when dried are reduced to one-fifth of that weight.

The small yield, the labor required, the care in culture and the difficulty of preserving the product in a good state renders saffron an expensive article—about 80c an ounce. On the seed-bearer of the flower there is a thread-like hook or fork, which at its upper head terminates in three thick, dark, orange-colored nerves or tissues; to save and collect these tissues the flowers are gathered in the fall, just as they are breaking, or a little before; they are plucked early in the morning, and these little masses are then pulled out with a considerable portion—about 1¼ in. of thread-like stem, to which they adhere. They are then dried over little charcoal fires or in the sun. It is this dried stigma, the trifid orange-colored tops of the central organ of the flower, that is the saffron of commerce. The remainder of the flower is useless.

Saffron as it generally comes to the trade consists of a large number of crooked and mixed-up threads, of an orange-red color; it has a peculiar, sharp, rooty and pungent smell, and a bitter balsam-like taste; that of a whitish yellow or blackish color is old and inferior. The great solubility of saffron prevents its use as a dye for fabrics, its place being taken by aniline dyes. Its coloring power is remarkable, a single grain rubbed to a fine powder with a little sugar will impart a distinct tint of yellow to 10 gals, of water; soaked in spirits or warm water it will yield three-fourths of its weight of a deep orange yellow coloring matter, which is perfectly wholesome, and if kept tightly corked will keep for some time. The chief uses of saffron are for flavoring and coloring confectionery and culinary articles; it is also used as a perfume and is given to birds during the moulting season. Spanish saffron is divided into five grades, according to the district in which it is cultivated. It is generally wrapped in tinfoil and then in white tissue paper and packed in tin boxes or strong cartons.

On account of its high price saffron is often counterfeited or adulterated with the petals of safflowers, African saffron, Meadow or wild saffron, marigold, arnica, etc. It is also loaded with glycerine, glucose, dyed vegetable filaments, honey, sulphate of soda, barium sulphate, etc., and exhausted saffron is sometimes re-colored with aniline dye. The stigma of genuine saffron immediately expands on being moistened with warm water, and its form is so characteristic that it cannot be mistaken for the flowerets of any of its adulterates.

Cake saffron is generally made from the dried flowers of the safflowers—a thistle-like plant of the aster family—or the florets of the saffron plants made into a paste with gum-water; it is used for dying and making rouge.

Turmeric

Turmeric is an East Indian plant (*curcuma longa*) of the ginger family, with the same properties as ginger, only not so powerful. It is also grown in Zanzibar, China and the Malayan archipelago. It is a stemless plant with dark green leaves varying from 6 in. to 24 in. long and 3 in. to 6 in. wide, flowers of a dull yellow color and a tuberous root varying in thickness from

that of a quill to ½ in. in diameter and often a foot long, with joints or ring-like swellings at short intervals; of, a yellowish to orange color outside and sometimes white and sometimes orange color inside. They are classed as long or round tubes according to their shape. From the root is made a kind of arrowroot much relished by the natives of India to color their faces. In medicine it is used as a cordial or stomachic; as an anti-scorbutic, and for stimulating the digestive organs. In a fresh state it is given to expel intestinal worms and in diarrhoea. It is used in varnishes and ointments and as a dye for silks and woollens, but it is now chiefly employed in making Indian curries or pickles, mustard, compounds, pudding spices, chow-chow pickles. A kind growing in Bengal, called “Mango ginger,” from its resemblance to the mango, is used for the same purpose as ginger.

Turmeric paper is a bibulous paper, yellow from saturation with the extract of turmeric, used as a test for alkalis, by which it is turned brown or red. Turmeric is also made from the roots of the canna, a member of the same family of plants cultivated at Sierra Leone.

Turmeric is adulterated with yellow ocher and carbonate of soda. Turmeric is insoluble in cold water, only partly soluble in boiling water, but is quite soluble in alcohol, forming beautiful yellow crystals.

Nasturtium

The flower buds and fruits of the common garden nasturtium are often used as a spice after being ground and dried; they are also pickled like capers and used on fish, meats, etc. The name is derived from nausea, nose, and tortus, twist, from the effects of its pungent smell or taste.

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Punctuation, hyphenation, and spelling were made consistent when a predominant preference was found in the original book; otherwise they were not changed.

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A Table of Contents was added by the Transcriber.

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Page [28](#): “the races of hands” probably should be “the races or hands”.

Page [32](#): “musterial” probably should be “material” or “materials”.

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